



KAVOSH ROBOTICS · U15 CAMP

MISSION IMPOSSIBLE

TRAINING CAMP · DAY 2

Today you will solve **2 robot missions** and **1 hands-on mission**. This time, your robot has to make decisions **live** — nothing is fully planned in advance.

KAVOSH ROBOTICS — U15 MISSION IMPOSSIBLE CAMP

SECRET ORDERS

Command can't radio you – the enemy is listening. Instead, coded markers have been left along your patrol route. Each colored marker is an order your robot must obey the moment it sees it – and you won't know which orders are coming until your robot is already moving.

YOU'LL NEED

- SPIKE robot (down color sensor)
- Taped line (~150 cm)
- Color markers (green / yellow / red)

THE CODE – LEARN THIS BEFORE YOU BUILD!

GREEN Keep going – no change.

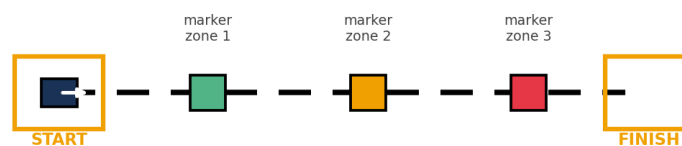
YELLOW Stop for 3 seconds, beep once, then continue.

RED Back up a little, then continue forward.

HOW IT WORKS

- 1 Your coach secretly places one marker in each of the three marker zones – any mix of green, yellow, and red. You will not know which colors until your robot sees them.
- 2 Press start – your robot follows the taped line.
- 3 The instant your robot crosses a marker, it must obey that color's order right away.
- 4 After obeying, your robot must find and follow the line again, all the way to each next marker.
- 5 Reach the Finish Box at the end of the line.
- 6 You have 2 minutes. You get 2 tries – your best try counts.

SECRET ORDERS – example field (top-down view)



coach picks any color for each zone — you find out live, one at a time

HOW YOU SCORE

What happens	Points
Each marker obeyed correctly (up to 3)	25 each
Robot reaches the Finish Box	+20
All 3 markers correct AND finished	+15

Best possible score: **110 points**

THE SCANNER

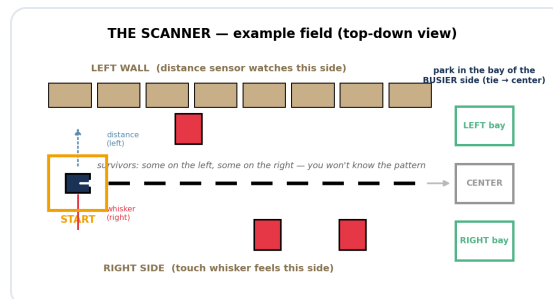
Survivors are trapped along both sides of the collapsed corridor — some on the left, some on the right. Your robot must drive through once, sense both sides at the same time, remember how many were on each side, and then report which side had more. Two sensors, two sides, one memory.

YOU'LL NEED

- SPIKE robot: LEFT distance sensor + RIGHT touch whisker
- Down color sensor (line following)
- Corridor with a left wall + decision bays

HOW IT WORKS

- 1 Your coach places 3 survivors along the corridor — each one on either the left or the right side. You won't know how many are on each side, or in what order.
- 2 Press start — your robot follows the line down the corridor, watching both sides at once: the distance sensor feels for survivors on the left, the touch whisker feels for survivors on the right.
- 3 Each time it finds a survivor, your robot must signal the side right away — 1 beep for a LEFT survivor, 2 beeps for a RIGHT survivor — and keep count of each side in its memory.
- 4 It keeps driving to the end of the corridor without leaving the line.
- 5 At the end, your robot must decide: drive into the LEFT bay if it found more survivors on the left, the RIGHT bay if it found more on the right, or the CENTER bay if it was a tie.
- 6 You have 2 minutes. You get 2 tries — your best try counts.



HOW YOU SCORE

What happens	Points
Each survivor found + correct side beep (up to 3)	20 each
Robot reaches the end without leaving the line	+15
Robot parks in the correct decision bay	+25

Best possible score: **100 points**

THE CRANE GANG

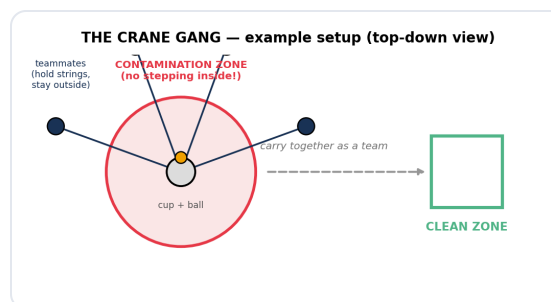
A container of medicine sits in the contamination zone. Anyone who touches it — or steps inside the zone — is compromised. Your whole team must work together to lift it, carry it across the room, and set it down safely, using only the rig you build.

YOU'LL NEED

- 1 rubber band
- 4 pieces of string (~1.5 m)
- Tape + scissors
- 1 cup + 1 ping-pong ball

HOW IT WORKS

- 1 Using only your allowed materials, build a rig that can grip and lift the cup without anyone touching it.
- 2 Reach over the edge of the contamination circle — never step inside it — to attach your rig to the cup. The ping-pong ball must stay balanced on top.
- 3 Lift the cup together as a team and carry it across to the Clean Zone. Every team member must be holding part of the rig during the carry.
- 4 Set the cup down inside the Clean Zone.
- 5 Just before each run, your coach will give you one surprise rule to follow — you'll only find out what it is right before you start.
- 6 You get 3 tries, 3 minutes each. Your best try counts.



HOW YOU SCORE

What happens	Points
Cup set down fully inside the Clean Zone	50
Cup set down touching the Clean Zone	30 instead
Cup carried at least halfway before a drop	15 instead
Ball never fell off (or no spill)	+30
Followed the surprise rule the whole time	+20

Best possible score: **100 points**

Day 2 – Quick Summary

Mission	Type	Difficulty	Main Skill
Secret Orders	Robot	Hard-Medium	React to signals while moving
The Scanner	Robot	Hard	Two-sided sensing + remember & decide
The Crane Gang	Hands-on	Hard	Teamwork under a surprise rule

Coach's plan for the day: Secret Orders (morning) → The Crane Gang (after lunch) → The Scanner (afternoon).